Feature importance of the 15:30 forecast (16:00 bar)

MDI, permutation importance on the causal held-out tail, SHAP — trees and the per-bar linear models; 1,469 forecasts
2018-06-25 .. 2024-04-30, 147 refits

Tree settings (read from the refit files). LightGBM: lgbm_opt_00 (392 rounds, 20 leaves, learning rate 0.0130, min child samples 8; the top entry of the 1998-2002 dev-span search); XGBoost: xgb_opt_00 (367 rounds, depth 5, learning rate 0.0123, min child weight 4.18; the top entry of the 1998-2002 dev-span search (since 2026-10-07; was the 7th-ranked xgb_opt_06)); random forest: trial 255 (485 trees, depth 20, min leaf 1, 90% of the features at a split, bootstrap of 41% of the rows; the top entry of the forest's 400-trial 1998-2002 dev-span search (since 2026-10-08; was the library defaults)). Leaf minimums are scaled from the pooled 24,000-row window to the 2,000-row window as in the per-bar spec.

Setup. Models: per-bar ridge and lasso (coefficients re-solved every session, penalty re-chosen causally every 250 sessions) and the untuned per-bar LightGBM, XGBoost and random forest (refit every 10 sessions; settings below). Every fit uses the 2000 sessions before it. A refit's *held-out tail* is the ≤ 10 sessions its model forecasts, up to the next refit: never in its fit, so a loss change measured there is out of sample and causal. Inputs: all features. Library: scikit-learn 1.9.0 (random forest), LightGBM 4.6.0, XGBoost 3.2.0, shap 0.51.0; the permutation and drop-column loops are written out, because a library permutation routine scores one fitted model on one test set, not 147 walk-forward models on their own tails.

Measures. (1) *MDI / gain*: LightGBM gain, XGBoost total gain, random-forest impurity decrease; share per refit, averaged. (2) *Permutation importance* on the tail: a unit's values are replaced with rows drawn from the refit's own training window (past data only; P2) and the change in loss is measured — QLIKE of the plain back-transform $\hat{y}^2\times$ diurnal baseline against the bar's realized variance (no recalibration) for the trees, squared error in the fit space for every model (the linear models' QLIKE permutation is not shown: moving an extreme input onto another row can push an unbounded linear forecast through zero); 10 draws per unit and refit, the same draws for all five models. A unit is a column or a *series* (all columns of one input: lags, flags, gates); a series' columns move together. Intervals: 95% circular-block bootstrap over refits (block 6). (3) *SHAP*: TreeSHAP (path-dependent) on the same tail rows; for the linear models $\beta_j(x_j - \bar{x}_j)$ with the window mean as background. Linear extras: $|\beta \cdot sd|$ and drop-column importance (re-solve without the unit, penalty held).

Key. Every exogenous input is divided by its time-of-day baseline, and the HAR columns `har_ma_*` are trailing means of the adjusted target, so no name carries a prefix. `har_ma_*` realized variance of the bar (the target's own HAR ladder, means over the last 1, 5, 25, 125, 625, 3125 bars; the per-bar design no longer carries the session-edge interactions `har_ma_*_x_open/close`); `sumabsret_ma_*` absolute return; `sumret_ma_*` signed return; `sumret3/4_ma_*` 3rd/4th-power returns; `sumpret2_ma_*` upside squared returns; `sumbipow_ma_*` bipower variation; `sumautocov_ma_*` return autocovariance; `sumvolume_ma_*` ES volume; `numobs_ma_*` ES prints per bar; `vix/vvix/vix3m_ma_*` the Cboe indices; `fomc_*` FOMC flags and distances; `*_avail/active_ma_*` is-present / is-nonzero flags; calendar = `DOW_*`, `is_*`, `hour`, `days_to_opex`. all features adds the constituent cross-section (`*_ewstock`, `*_vwstock`), Cboe volume (`voldemand_*`) and StockTwits (`stocktwits_*`).

Table 1: The design is float throughout, but not all continuous: distinct values per column in each 2000-session window (median over the 147 windows) — what MDI's cardinality bias keys on.

bucket	columns	constant	binary	3–25	26–500	>500	series
all features	628	137	40	93	91	267	49

1. How much do the inputs below the leader matter?

Table 2: Permutation importance P2 at the series level, all features (the change in the tail loss when a series is replaced with rows drawn from the refit's training window; mean over the refits, 95% interval over refits). Columns: `har_ma_*`'s own value ($\times 10^3$); of the other series, those whose interval excludes zero, those whose lower bound clears the upper bound of a pure-noise probe column (trees, QLIKE); their summed importance and the largest single series, both as % of `har_ma_*`'s.

model	loss	`har_ma_*`	interval > 0	above noise	their sum	largest
ridge	sq. error	85	20	–	77.2%	15.4% (`sumabsret_ma_*`)
lasso	sq. error	170	20	–	24.2%	5.7% (`sumabsret_vwstock_ma_*`)
LightGBM	QLIKE	249	8	6	10.1%	3.2% (`calendar`)
LightGBM	sq. error	127	4	–	4.2%	2.0% (`calendar`)
XGBoost	QLIKE	226	8	5	10.6%	3.1% (`calendar`)
XGBoost	sq. error	114	5	–	4.4%	1.9% (`calendar`)
random forest	QLIKE	340	7	4	3.8%	0.9% (`sumvolume_ma_*`)
random forest	sq. error	158	4	–	1.5%	0.9% (`sumvolume_ma_*`)

Reading. For the trees, everything below the HAR ladder whose interval excludes zero is together worth 4–11% of the ladder's permutation importance on QLIKE (1.5–4.4% on squared error), and the largest single input at most 3.2%. For the linear models the credit is spread more widely (ridge: 20 series, together 77% of the ladder's; lasso 24%): the penalty shrinks the ladder and correlated inputs share its weight. So for the trees the ranking below the leader orders inputs that are small in absolute terms, and the cardinality bias below matters only for that ordering.

Cardinality bias. MDI and TreeSHAP favour many-valued columns; the tables below show how much, and why their rankings below the leader are not a feature list.

Table 3: Spearman rank correlation, over the non-constant columns, between a column's importance and its number of distinct values. MDI and TreeSHAP track cardinality; permutation importance does not.

bucket	model	MDI	SHAP	perm P2 QL	perm P2 MSE
all features	LightGBM	0.82	0.79	-0.04	-0.05
all features	XGBoost	0.82	0.79	0.04	0.00
all features	random forest	0.82	0.80	0.13	-0.02

Table 4: Noise probes: on every third refit a second fit appends three pure-noise columns (continuous; 20 values; binary). Median rank among all columns (1 = most important) and, in brackets, the share of the real columns the model uses that rank below the probe; permutation P2 of the continuous probe (ΔQLIKE $\times 10^3$, 95% interval) against `har_ma_*`'s.

bucket	model	cols	MDI rank	SHAP rank	20-val MDI	binary MDI	perm P2 (cont.)	`har_ma_*`
all features	LightGBM	631	73 (77%)	82 (70%)	138	315	−0.07 [-0.40, +0.25]	249
all features	XGBoost	631	88 (74%)	110 (64%)	156	310	+0.04 [-0.33, +0.37]	226
all features	random forest	631	52 (85%)	96 (72%)	122	308	+0.09 [-0.07, +0.27]	340

Reading. A column of pure noise with a new value on every row is ranked by MDI above 77%, 74%, 85% (LightGBM, XGBoost, forest) of the real all-features columns the model uses; the binary noise column sits near the bottom. The same noise gets a permutation importance of at most 0.09×10^{-3} in absolute value (3 of 3 intervals include zero), against at least 0.23 for `har_ma_*`. TreeSHAP, which splits the fitted trees, inherits part of the bias. Over the real columns, MDI and SHAP correlate 0.82–0.80 with the number of distinct values, permutation -0.04–0.13. So below the leader, the MDI ranking of this design is partly a ranking by cardinality; the permutation ranking is the one to use for selection. At the series level: `har_ma_*` is the leader of 12 of 12 tree bucket x model x measure rows. The cardinality bias matters for everything below the leader. Series that MDI puts in the top 5 but permutation P2 ranks outside the top 10 (permutation rank in brackets): all features LightGBM: `sellturnover_vwstock_ma_*` (36); all features XGBoost: `sellturnover_vwstock_ma_*` (33); all features random forest: none.

2. The top inputs by measure

Table 5: All-features design, series level: top three per model and measure (share for MDI / $|\beta \cdot sd|$ / SHAP; change as % of the model's tail loss for permutation and drop-column).

model	measure	top three
ridge	$\|\beta \cdot sd\|$	`har_ma_*` 11%; `sumabsret_ma_*` 5%; `sumabsret_vwstock_ma_*` 5%
ridge	perm. P2, ΔMSE	`har_ma_*` +142%; `sumabsret_ma_*` +22%; `sumret4_ma_*` +18%
ridge	drop-column, ΔMSE	`har_ma_*` +17%; `calendar` +2%; `sumvolume_ma_*` +2%
ridge	SHAP mean $\|\phi\|$	`har_ma_*` 15%; `sumabsret_ma_*` 6%; `sumabsret_vwstock_ma_*` 5%
lasso	$\|\beta \cdot sd\|$	`har_ma_*` 46%; `sumabsret_vwstock_ma_*` 10%; `sumret_ma_*` 9%
lasso	perm. P2, ΔMSE	`har_ma_*` +294%; `sumabsret_vwstock_ma_*` +17%; `vix_ma_*` +15%
lasso	drop-column, ΔMSE	`har_ma_*` +27%; `sumret_ma_*` +2%; `sumabsret_vwstock_ma_*` +1%
lasso	SHAP mean $\|\phi\|$	`har_ma_*` 46%; `sumabsret_vwstock_ma_*` 10%; `sumret_ma_*` 9%
LightGBM	MDI / gain	`har_ma_*` 52%; `sumabsret_ma_*` 6%; `sumret_ma_*` 5%
LightGBM	perm. P2, ΔQLIKE	`har_ma_*` +200%; `calendar` +6%; `sumret_ma_*` +3%
LightGBM	perm. P2, ΔMSE	`har_ma_*` +204%; `calendar` +4%; `sumvolume_ma_*` +3%
LightGBM	SHAP mean $\|\phi\|$	`har_ma_*` 44%; `sumret_ma_*` 6%; `sumvolume_ma_*` 5%
XGBoost	MDI / gain	`har_ma_*` 54%; `sumabsret_ma_*` 7%; `sumret_ma_*` 5%
XGBoost	perm. P2, ΔQLIKE	`har_ma_*` +181%; `calendar` +6%; `sumvolume_ma_*` +4%
XGBoost	perm. P2, ΔMSE	`har_ma_*` +175%; `calendar` +3%; `sumvolume_ma_*` +3%
XGBoost	SHAP mean $\|\phi\|$	`har_ma_*` 47%; `sumret_ma_*` 7%; `sumabsret_ma_*` 5%
random forest	MDI / gain	`har_ma_*` 59%; `sumabsret_ma_*` 5%; `sumret_ma_*` 3%
random forest	perm. P2, ΔQLIKE	`har_ma_*` +271%; `sumvolume_ma_*` +2%; `calendar` +2%
random forest	perm. P2, ΔMSE	`har_ma_*` +249%; `sumvolume_ma_*` +2%; `sumabsret_ma_*` +2%
random forest	SHAP mean $\|\phi\|$	`har_ma_*` 61%; `sumabsret_ma_*` 5%; `sumret_ma_*` 3%

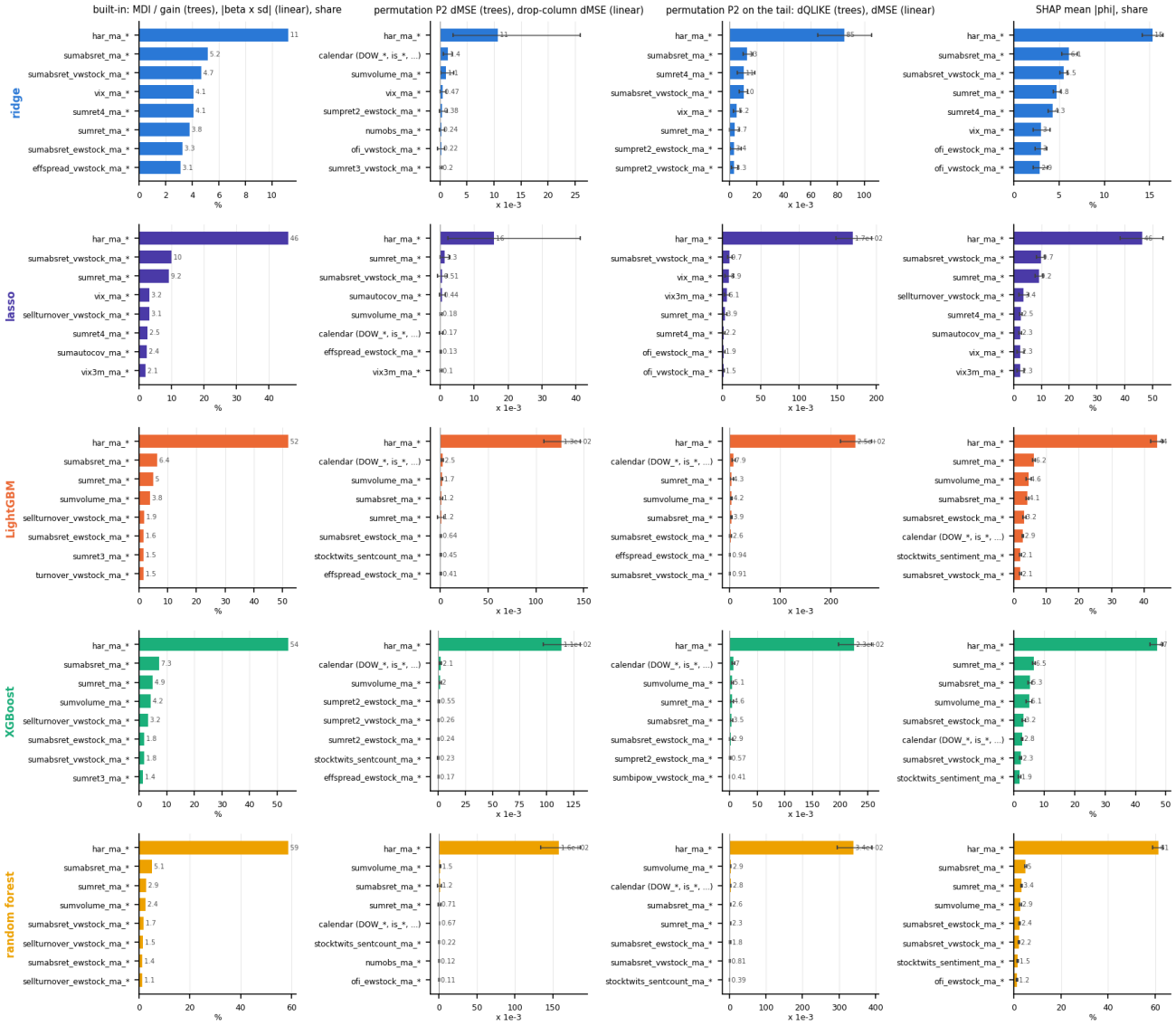


Figure 1: All features: ranked bars per measure (series level; top 8). Rows: models; columns: the built-in measure (MDI / gain for the trees, $|\beta \cdot sd|$ for the linear models), permutation P2 ΔMSE (trees) / drop-column ΔMSE (linear), permutation P2 on the tail (ΔQLIKE for the trees, ΔMSE for the linear models), SHAP. Bars with 95% intervals over refits. Linear permutation / drop-column in ΔMSE (see Section 6).

3. Ranking agreement

Table 6: Within a tree model: Spearman between measures (series / column level).

bucket	level	model	MDI∼SHAP	MDI∼P2	SHAP∼P2	P2 QL∼MSE
all features	series	LightGBM	0.87	0.21	0.27	0.72
all features	series	XGBoost	0.87	0.16	0.20	0.36
all features	series	random forest	0.88	0.09	0.29	0.64
all features	column	LightGBM	0.97	0.06	0.05	0.60
all features	column	XGBoost	0.97	0.10	0.12	0.42
all features	column	random forest	0.97	0.21	0.22	0.50

Table 7: Within a linear model: Spearman between measures (permutation and drop-column in squared error).

bucket	level	model	$\|\beta sd\|$ ∼SHAP	$\|\beta sd\|$ ∼P2	SHAP∼P2	P2∼drop
all features	series	ridge	0.83	0.59	0.73	0.38
all features	series	lasso	0.95	0.88	0.86	0.13
all features	column	ridge	0.99	0.42	0.42	0.16
all features	column	lasso	0.99	0.40	0.38	0.13

Table 8: Across models, same measure: range of Spearman over tree–tree pairs, tree–linear pairs, and ridge–lasso (QLIKE permutation: trees only).

bucket	level	SHAP			perm P2 ΔQLIKE	perm P2 ΔMSE		
		tree	tree–lin	r–l	tree	tree	tree–lin	r–l
all features	series	0.84–0.97	0.33–0.49	0.86	0.65–0.70	0.44–0.68	-0.08–0.49	0.69
all features	column	0.96–0.99	0.67–0.88	0.78	0.25–0.34	0.24–0.33	-0.05–0.08	0.43

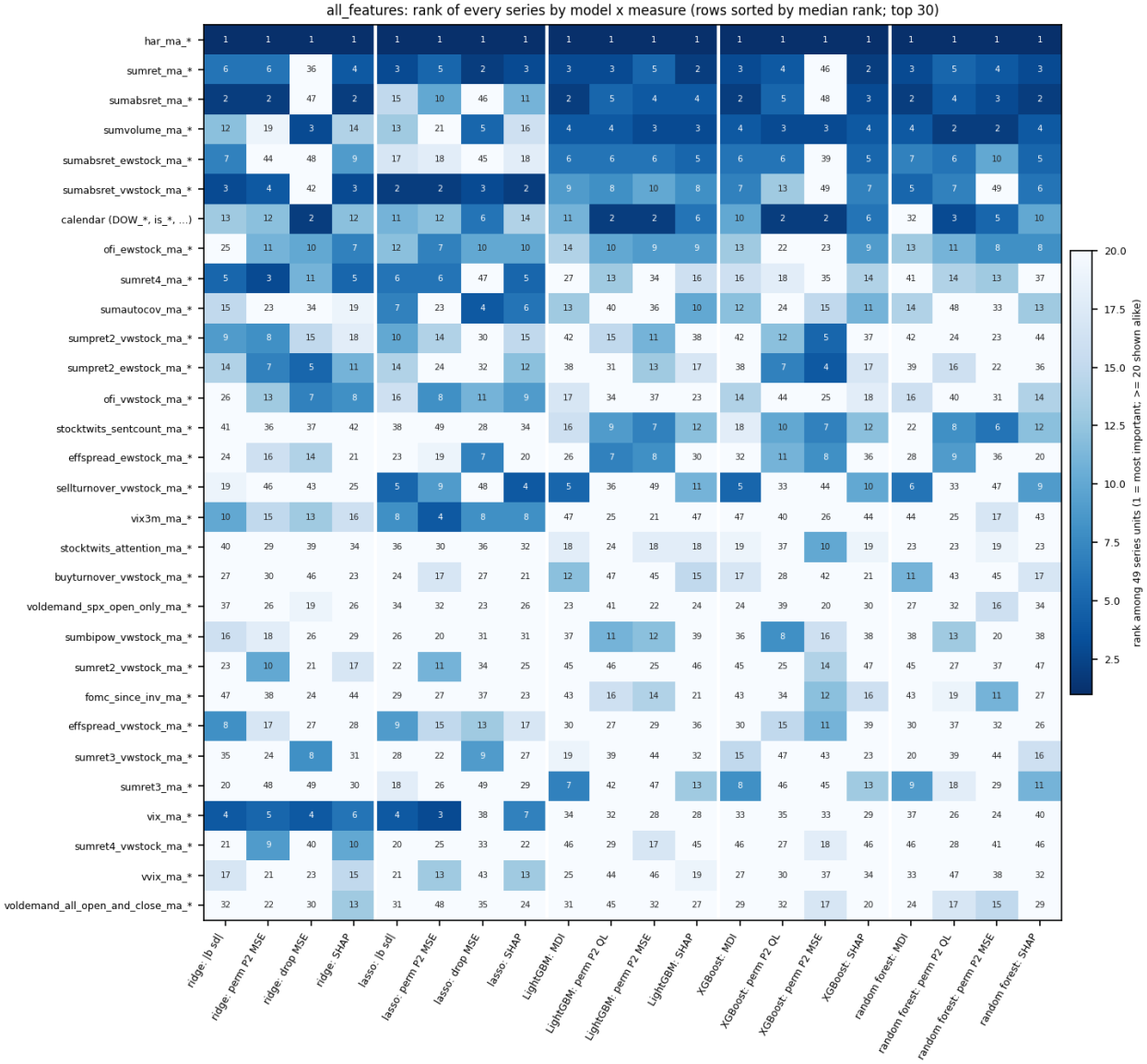


Figure 2: All features: rank of every series under every model × measure (1 = most important); measures: MDI / $|\beta \cdot sd|$, permutation P2 (ΔQLIKE trees only, ΔMSE), drop-column (linear), SHAP.

4. Stability over refits and regimes

Table 9: Top-5 stability (series level): share of the 147 refits in which `har_ma_*` is in the refit's top 5; number of series in the top 5 in at least half the refits; number ever in the top 5. Measures: built-in (MDI or $|\beta sd|$), permutation P2 (ΔQLIKE trees, ΔMSE linear), SHAP.

bucket	model	built-in			perm P2			SHAP		
		har	$\geq$50%	ever	har	$\geq$50%	ever	har	$\geq$50%	ever
all features	ridge	100%	5	11	98%	3	44	99%	3	27
all features	lasso	100%	4	12	100%	3	30	100%	3	24
all features	LightGBM	100%	5	10	99%	2	47	100%	4	28
all features	XGBoost	100%	5	10	99%	3	48	100%	4	23
all features	random forest	100%	5	10	99%	2	47	100%	4	24

Table 10: Regimes: Spearman between the ranking within a stratum and the full-sample ranking (series level): median (minimum) over the calendar years (2018: 13, 2019: 26, 2020: 25, 2021: 25, 2022: 25, 2023: 25, 2024: 8 refits) and over the causal VIX terciles (VIX at 15:30 against the 1/3, 2/3 quantiles of the previous 2000 sessions' 15:30 VIX: low 28, mid 38, high 81 refits); years in which the full-sample leader also leads.

bucket	model	built-in			perm P2			SHAP		
		year	VIX	lead	year	VIX	lead	year	VIX	lead
all features	ridge	0.93 (0.84)	0.99 (0.95)	7/7	0.67 (0.50)	0.78 (0.60)	7/7	0.76 (0.54)	0.92 (0.83)	7/7
all features	lasso	0.60 (0.47)	0.99 (0.84)	7/7	0.39 (0.15)	0.72 (0.31)	7/7	0.64 (0.42)	0.97 (0.83)	7/7
all features	LightGBM	0.94 (0.87)	0.99 (0.98)	7/7	0.60 (0.31)	0.66 (0.61)	7/7	0.83 (0.73)	0.93 (0.93)	7/7
all features	XGBoost	0.94 (0.91)	0.99 (0.98)	7/7	0.51 (0.40)	0.74 (0.36)	7/7	0.87 (0.68)	0.95 (0.93)	7/7
all features	random forest	0.98 (0.89)	1.00 (0.98)	7/7	0.50 (0.30)	0.56 (0.49)	7/7	0.91 (0.75)	0.98 (0.96)	7/7

5. Tuned trees

Table 11: Causally tuned trees (their own extracts: gain per refit and TreeSHAP rows at 16:00): `har_ma_*` share tuned (untuned), rank correlation tuned vs untuned over the series, and the next three series by tuned SHAP. Permutation was not computed for the tuned trees (their per-refit configurations would have to be refitted).

bucket	model	MDI har	SHAP har	ρ MDI	ρ SHAP	next by tuned SHAP
all features	LightGBM	51% (52%)	40% (44%)	0.90	0.90	`sumabsret`, `sumvolume`, `sumret`
all features	XGBoost	50% (54%)	41% (47%)	0.93	0.90	`sumvolume`, `sumabsret`, `sumret`
all features	random forest	44% (59%)	41% (61%)	0.70	0.72	`sumabsret`, `sumvolume`, `sumabsret_vwstock`

6. Importance over time

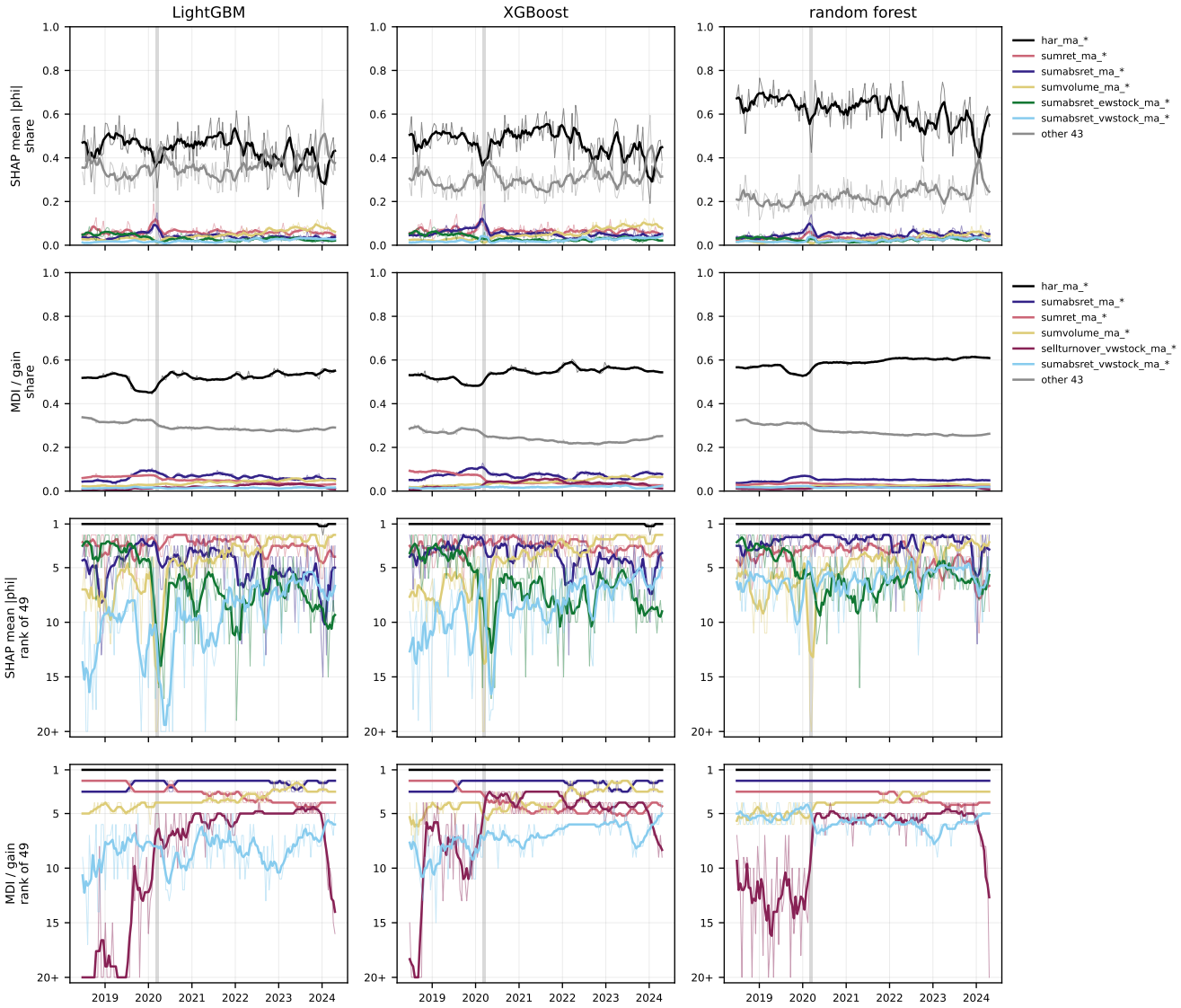


Figure 3: Importance over time, all features, series level (a series = the sum of its columns), one column per tree: LightGBM (lgbm_opt_00), XGBoost (xgb_opt_00), random forest (trial 255). Rows 1–2: each series' share of the refit's total importance, SHAP mean $|\phi|$ on the refit's held-out tail and MDI / gain of the fitted model; rows 3–4: the same series' rank among the 49 series (1 = most important; ranks beyond 20 drawn at 20). Lines: `har_ma_*` and the next 5 series by overall share (mean over the refits and the three trees), the rest summed as 'other'. x axis: the start date of each of the 147 refits' tails. Thin lines: one point per refit — the trees are refit every 10 sessions, and each point is that refit's model on its own held-out tail (the ≤ 10 sessions it forecasts before the next refit): SHAP averaged over those tail rows, MDI / gain of the fitted model; thick lines: 5-refit centred rolling mean. Grey band: the 2020 crash tails (refits 42–43, tails starting 2020-02-28 and 2020-03-13). Mean `har_ma_*` share and rank-1 count: LightGBM 44% SHAP / 52% MDI, rank 1 in 146 / 147 of 147 refits; XGBoost 47% SHAP / 54% MDI, rank 1 in 146 / 147 of 147 refits; random forest 61% SHAP / 59% MDI, rank 1 in 147 / 147 of 147 refits. SHAP and MDI shares are split credit: `har_ma_*` shares its credit with every input correlated with the HAR ladder, and MDI and SHAP inflate continuous, many-valued columns (the noise-probe and cardinality checks above); the shares below `har_ma_*` are therefore not a ranked feature list.

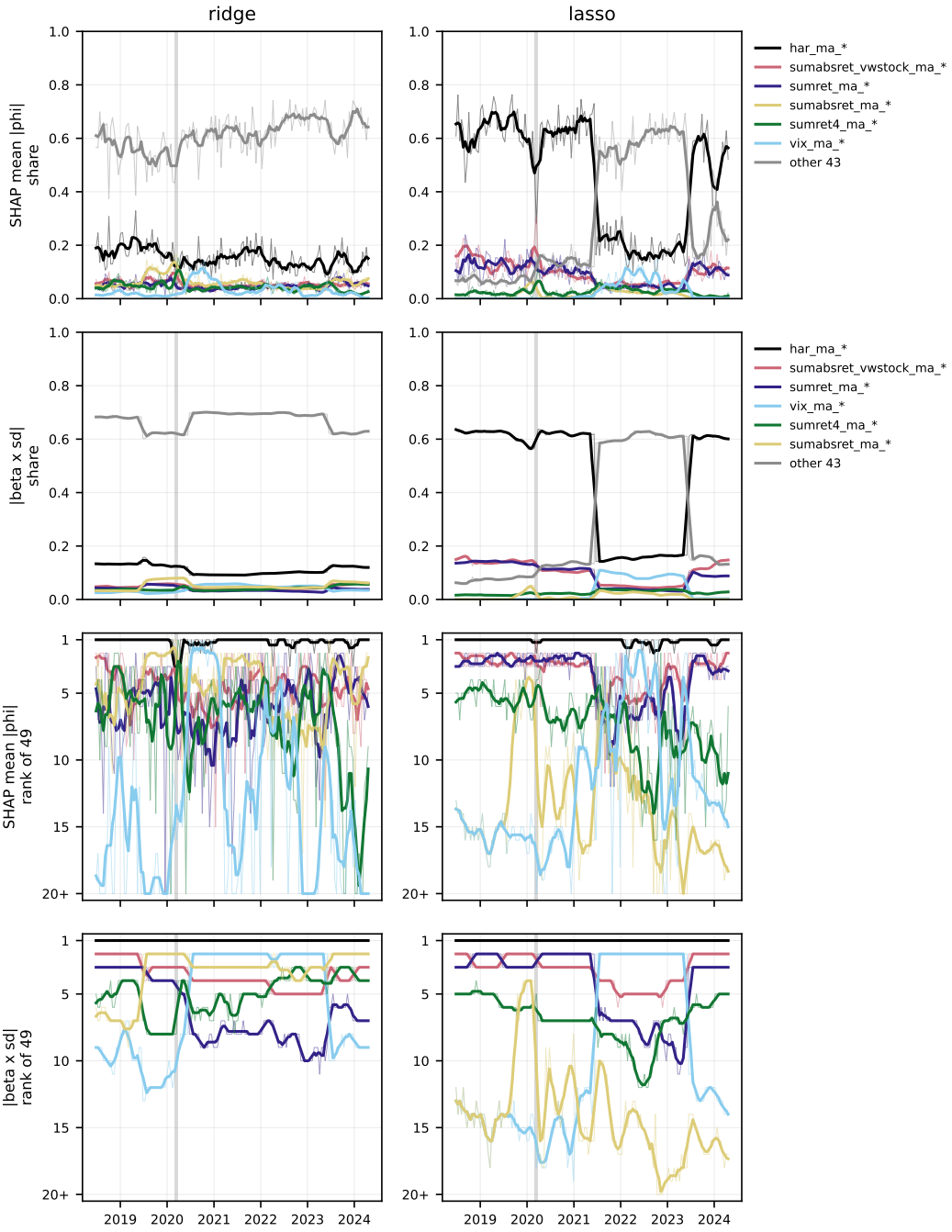


Figure 4: The same for the per-bar linear models (ridge and lasso, as in Setup), one column per model. Rows 1–2: each series' share of the refit's SHAP mean $|\phi|$ ($\phi_j = \beta_j(x_j - \bar{x}_j)$, window mean as background) on the refit's held-out tail, and of $|\beta \cdot sd|$ (coefficient times the column's standard deviation in the refit's 2000-session window, summed over a series' columns); rows 3–4: ranks among the 49 series. The same 147 refit points and tails as the trees (one point per 10 sessions), the same smoothing and grey crash band. Mean `har_ma_*` share and rank-1 count: ridge 15% SHAP / 11% $|\beta \cdot sd|$, rank 1 in 132 / 147 of 147 refits; lasso 46% SHAP / 46% $|\beta \cdot sd|$, rank 1 in 137 / 147 of 147 refits. A linear model's credit is split across correlated columns differently from a tree's (the penalty spreads or concentrates it), so the shares compare over time within one model, not across the two figures.